

Chapter 2

Series

In
this
chap-
ter
we
build
on
the
con-
cepts
we
de-
vel-
oped
in
the
se-
quences
chap-
ter
and
ap-
ply
them
to
se-
ries.
We
will
look
at
the
con-
cept
of
the
con-
ver-
gence
of

CHAPTER

2.

SERIES

an

in-

fi-

nite

se-

ries

and

re-

sults

that

give

var-

i-

ous

con-

di-

tions

for

their

con-

ver-

gence

or

di-

ver-

gence.

Two

im-

por-

tant

uses

of

se-

ries

are

in

func-

tions

ex-

pan-

sions

(such

as

Tay-

lor

se-

ries)

and

the

def-

i-

ni-

tion

of

the

Rie-

2.1.
BA-
SIC
DEFINITIONS
in-
te-
gral.

2.1 Basic def- i- ni- tions

A
se-
ries
is
the
name
given
to
the
pro-
cess
where
we
take
a
se-
quence
of
terms
and
add
them
all
to-
gether
term
by
term.

Definition

2.1.

A
*se-
ries*
is
the
sum
of
the
terms
of
a
se-
quence.

CHAPTER

2.

SERIES

We

can

rep-

re-

sent

se-

ries

us-

ing

ad-

di-

tion

and

sub-

trac-

tion

signs

as

in

$$1-2+3-4+5-6+7-8+9-10,$$

or

us-

ing

the

sigma

no-

ta-

tion

(Σ

is

the

cap-

i-

tal

Greek

let-

ter

sigma)

as

in

$$\sum_{k=1}^{10} (-1)^{k-1} k.$$

A

se-

ries

can

con-

tain

a

fi-

nite

num-

ber

2.1.

BA-

SIC

91 DEFINITIONS

terms

as

in

$\sum_{n=1}^N x_n$ or equivalently $x_1+x_2+\cdots+x_N$,

or

an

in-

fi-

nite

num-

ber

of

terms

as

in,

$\sum_{n=1}^{\infty} x_n$ or equivalently $x_1+x_2+\ldots$

If

a

se-

ries

con-

tains

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nite

num-

ber

of

terms

then

the

se-

ries

will

sum

to

a

fi-

nite

value

(as-

sum-

ing,

as

we

are

here,

that

the

CHAPTER

2.

SERIES

terms

of

the

se-

ries

are

all

fi-

nite

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ti-

ties).

If

a

se-

ries

has

an

in-

fi-

nite

num-

ber

of

terms

then

the

se-

ries

may

or

may

not

sum

to

a

fi-

nite

value.

The

pre-

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ming

to

a

fi-

nite

value

is

given

2.1.
 BA-
 SIC
 DEFINITIONS

the
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 low-
 ing.
 First
 we
 de-
 fine
 what
 is
 meant
 by
 the
 par-
 tial
 sums
 of
 an
 in-
 fi-
 nite
 se-
 ries

Definition

2.2
 (Par-
 tial
 sums).
 Con-
 sider
 an
 in-
 fi-
 nite
 se-
 ries
 $\sum_{n=1}^{\infty} x_n$.

The
 k^{th} -
partial
sum,
 S_k ,
 of
 the
 se-
 ries
 is
 the
 fi-
 nite
 se-
 ries

$$S_k = \sum_{n=1}^k x_n = x_1 + x_2 + \cdots + x_k.$$

CHAPTER

2.

SERIES

2.2 Convergence

Definition

2.3

(Con-

ver-

gence

for

se-

ries).

The

se-

ries

$\sum_{n=1}^{\infty} x_n$

con-

verges

to

the

limit

(or

sum)

L

if

the

se-

quence

of

its

par-

tial

sums,

$\{S_k\}_{k=1}^{\infty}$,

is

a

con-

ver-

gent

se-

quence

and

$$\lim_{k \rightarrow \infty} S_k = L.$$

If

this

is

the

case

then

we

can

sim-

ply

write

$$\sum_{n=1}^{\infty} x_n = L.$$

2.2. CONVERGENCE

A
se-
ries
is
di-
ver-
gent
if
the
se-
quence
of
its
par-
tial
sums
is
a
di-
ver-
gent
se-
quence.

It
is
not
hard
to
think
of
ex-
am-
ples
of
di-
ver-
gent
se-
ries.

Example

2.1
(A
di-
ver-
gent
se-
ries).
Con-
sider
the
se-
ries

$$\sum_{m=1}^{\infty} m.$$

CHAPTER

2.

SERIES

This

se-

ries

is

clearly

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gent

as

the

par-

tial

sums

are

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with-

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bound,

and

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can-

not

form

a

con-

ver-

gent

se-

quence

(see

the-

o-

rem

??).

However

it

is

not

so

ob-

vi-

ous

that

con-

ver-

gent

se-

ries

ex-

ist

2.2. CONVERGENCE

(be-
yond
the
triv-
ial
ex-
am-
ple
of
a
se-
ries
where
the
terms
are
all
zero
af-
ter
some
point).

Example 2.2

(A
con-
ver-
gent
se-
ries).
Con-
sider
the
se-
ries

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m.$$

The
con-
ver-
gence
of
this
se-
ries
can
be
ap-
pre-
ci-
ated
when
we
in-

CHAPTER

2.

SERIES

ter-

pret

the

terms

as

mea-

sur-

ing

the

size

of

step

taken

along

a

line.

We

see

that

the

par-

tial

sums

of

the

se-

ries

get

closer

and

closer

to

1.

In-

deed

the

se-

ries

does

con-

verge

to

1.

This

can

be

proved

us-

ing

the

par-

tial

sum

for-

mula

for

2.2. CONVERGENCE

a
ge-
o-
met-
ric
se-
ries
and
mak-
ing
use
of
the
al-
ge-
bra
of
se-
quences
the-
o-
rem.

This
ex-
am-
ple
shows
that
it
is
pos-
si-
ble
to
have
an
in-
fi-
nite
num-
ber
of
terms
and
yet
for
them
still
to
sum
to
a
fi-
nite
value.

CHAPTER

2.

~~SE~~RIES

This happens because the terms are getting smaller and smaller, the terms themselves are converging to zero, when viewed as sequence. However this type of behaviour does not guarantee convergence for a series, as the next example shows.

Example

2.2. CONVERGENCE

2.3

(A
di-
ver-
gent
se-
ries:
The
Har-
monic
Se-
ries).
Con-
sider
the
se-
ries

$$\sum_{m=1}^{\infty} \frac{1}{m}.$$

The
terms
of
this
se-
ries
are
in-
deed
con-
verg-
ing
to
zero,
but
yet
the
se-
ries
does
not
con-
verge.
It
in-
creases
with-
out
any
up-
per
bound,
al-
beit
very
slowly.

CHAPTER

2.

SERIES

This

fact

can

be

proved

with

the

aid

of

a

group-

ing

ar-

gu-

ment.

Note

that

chang-

ing

any

fi-

nite

num-

ber

of

terms

of

a

se-

ries

will

not

af-

fect

whether

a

se-

ries

is

con-

ver-

gent

or

di-

ver-

gent

(al-

though

it

may

well

change

the

value

of

2.2. CONVERGENCE

its
sum)
as
con-
ver-
gence
for
a
se-
ries
de-
pends
on
its
'long
term'
be-
haviour.

For
se-
ries
that
con-
tain
a
mix-
ture
of
pos-
i-
tive
and
neg-
a-
tive
terms
it
is
help-
ful
to
be
able
to
re-
fer
to
the
con-
ver-
gence
of
the
re-
lated
se-

CHAPTER

2.

SERIES

ries

of

ab-

so-

lute

val-

ues.

Definition

2.4.

A

se-

ries

$$\sum_{m=1}^{\infty} a_m$$

is

ab-

so-

lutely

con-

ver-

gent

if

the

se-

ries

of

ab-

so-

lute

val-

ues,

$$\sum_{m=1}^{\infty} |a_m|,$$

is

a

con-

ver-

gent

se-

ries.

If

the

se-

ries

$$\sum_{m=1}^{\infty} a_m$$

is

con-

ver-

gent

but

the

se-

ries

of

ab-

so-

lute

2.2. CONVERGENCE

val-
ues,
 $\sum_{m=1}^{\infty} |a_m|$,
is
di-
ver-
gent,
then
the
orig-
i-
nal
se-
ries
is
said
to
be
*con-
di-
tion-
ally
con-
ver-
gent.*

The
im-
por-
tance
of
con-
di-
tional
con-
ver-
gence
is
given
by
the
fol-
low-
ing
re-
sult
which
we
shall
quote
with-
out
proof.

Theorem
2.1.
If

CHAPTER

2.

~~S~~ERIES

a

se-

ries

is

ab-

so-

lutely

con-

ver-

gent

then

it

is

con-

ver-

gent.

More-

over,

the

terms

of

an

ab-

so-

lutely

con-

ver-

gent

se-

ries

can

be

re-

ar-

ranged

in

any

way

and

the

re-

sult-

ing

se-

ries

will

still

con-

verge

to

the

same

sum.

How-

ever

if

2.3.

A

SHORT

SEC-

TION

ON

TAILS

con-

di-

tion-

ally

con-

ver-

gent

then

re-

ar-

rang-

ing

its

terms

can

yield

a

se-

ries

which

sums

to

a

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fer-

ent

value.

Proof.

Omitted.

□

2.3 A

short

sec-

tion

on

tails

For

an

in-

fi-

nite

se-

ries

$$\sum_{m=1}^{\infty} a_m = a_1 + a_2 + a_3 + a_4 + \dots$$

we

note

CHAPTER

2.

SERIES

that

the

con-

ver-

gence

or

di-

ver-

gence

of

the

se-

ries

is

not

al-

tered

if

some

fi-

nite

col-

lec-

tion

of

the

terms

of

the

se-

ries

are

changed.

For

sup-

pose

that

we

change

a_1

to

$a_1 +$

δ_1

and

a_2

to

$a_2 +$

δ_2 .

If

the

orig-

i-

nal

se-

ries

is

2.3.

A

SHORT

CON-

VER-

TION

ON,

SE-

RIES

A

then

chang-

ing

these

val-

ues

will

mean

the

new

se-

ries

con-

verges

to

A+

$\delta_1 +$

δ_2 .

On

the

other

hand

if

the

se-

ries

is

di-

ver-

gent

it

is

be-

cause

the

se-

quence

of

par-

tial

sums,

$S_k =$

$\sum_{m=1}^k a_m,$

fails

to

con-

verge.

The

par-

tial

CHAPTER

2.

~~SERIES~~

sums,

S'_k ,

of

the

new

se-

ries,

for

$k \geq$

3,

will

sat-

isfy

$S'_k =$

$S_k +$

$\delta_1 +$

δ_2 ,

and

so

these

too

will

still

fail

to

con-

verge.

Both

of

these

ar-

gu-

ments

ex-

tend

to

any

fi-

nite

col-

lec-

tion

of

terms.

By

a

tail

of

the

se-

ries

$\sum_{m=1}^{\infty} a_m$

we

mean

any

se-

2.3.

A

SHORT

des-

SEC-

TION

of

the

ON

form

$\sum_{m=M}^{\infty} a_m,$

for

some

pos-

i-

tive

in-

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ger

$M,$

i.e.

the

orig-

i-

nal

se-

ries

less

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nite

col-

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of

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tial

terms.

If

we

can

prove

that

some

tail

of

the

se-

ries

is

con-

ver-

gent,

then

so

will

the

orig-

i-

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se-

ries

CHAPTER

2.

SERIES

be

con-

ver-

gent.

Sim-

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larly,

if

we

can

show

that

some

tail

of

the

se-

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is

di-

ver-

gent,

then

so

will

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orig-

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se-

ries

be

di-

ver-

gent.

These

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in

some

of

the

proofs

of

the

se-

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con-

ver-

gence

tests

later

in

the

chap-

2.4.
SOME
SE-
RIES
CON-
VER-
GENCE
PROPERTIES
of
con-
ver-
gence
prop-
er-
ties

Theorem

2.2

(Lin-
ear
com-
bi-
na-
tions
of
con-
ver-
gent
se-
ries).

*Sup-
pose
that*

*we
have*

two

con-

ver-

gent

se-

ries

$\sum_{m=1}^{\infty} a_m$

and

$\sum_{m=1}^{\infty} b_m$

such

that

$$\sum_{m=1}^{\infty} a_m = A, \text{ and } \sum_{m=1}^{\infty} b_m = B.$$

Then

for

any

pair

$\alpha, \beta \in$

$\mathbb{R}$

we

have

a

CHAPTER

2.

SERIES

~~SE~~
new

se-

ries

$$\sum_{m=1}^{\infty} (\alpha a_m + \beta b_m),$$

which

is

con-

ver-

gent

and

sat-

is-

fies

$$\sum_{m=1}^{\infty} (\alpha a_m + \beta b_m) = \alpha A + \beta B.$$

Proof.

This

can

be

proved

us-

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lin-

ear-

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prop-

erty

for

con-

ver-

gence

of

se-

quences.

We

do

not

need

to

de-

scend

to

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ar-

gu-

ments

but

just

use

the

2.4.
SOME
SE-
RIES
CON-
VER-
GENCE
PROPERTIES

se-
ries
con-
ver-
gence
in
terms
of
the
par-
tial
sums.

Letting

A_k
and
 B_k
de-
note
the
par-
tial
sums,

i.e.

$$A_k = \sum_{n=1}^k a_n$$

and

$$B_k = \sum_{n=1}^k b_n$$

we
make
the
fol-
low-
ing
de-
duc-
tions.

Direction of presentation

Why was the presentation reversed
Here? ~~But we started with B_k~~
 ~~$\alpha A + \beta B$ and worked our~~
way towards $\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n)$, by linearity of limits for sequences

$$\begin{aligned} &= \lim_{k \rightarrow \infty} \left(\alpha \sum_{n=1}^k a_n + \beta \sum_{n=1}^k b_n \right) \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k (\alpha a_n + \beta b_n), \text{ manipulation of sum} \\ &= \sum_{n=1}^{\infty} (\alpha a_n + \beta b_n), \text{ by def. of series convergence.} \end{aligned}$$

2.5 Convergence tests

We will introduce various tests that can be used to assess the convergence / divergence of a series.

re of a statement $A \Rightarrow$
nt $\neg B \Rightarrow \neg A$. These
e *logically equivalent*,
ave the same truth
true then so is the
sult one is trying to
rm of an implication,
the contrapositive of
e easier to prove di-

Theorem
2.3
(Gen-
eral
term
test).
If
a
se-
quence
has
a
non-
zero
limit
(*or*
no
limit
at
all)
then
the

2.5.
CON-
VER-
GENCE
TESTS
ci-
ated
se-
ries
will
di-
verge.
Equiv-
a-
lently,
by
form-
ing
the
contra-
positive
we
can
quote
this
test
as:
If
a
se-
ries
con-
verges
then
the
as-
so-
ci-
ated
se-
quence
of
terms
con-
verges
to
zero.
Sym-
bol-
i-
cally
these
two
state-
ments
are

•

CHAPTER

2.

SERIES

$$\lim_{m \rightarrow \infty} a_m \neq 0 \Rightarrow \sum_{m=1}^{\infty} a_m \text{ diverges,}$$

•

$$\sum_{m=1}^{\infty} a_m \text{ converges} \Rightarrow \lim_{m \rightarrow \infty} a_m = 0.$$

Proof.

Suppose

that

$$\sum_{n=1}^{\infty} a_n$$

con-

verges

and

let

$$A_k$$

de-

note

the

par-

tial

sum

as

be-

fore,

i.e.

$$A_k \rightarrow$$

a

as

$$k \rightarrow$$

∞ .

Let

$$\epsilon >$$

0

be

given.

There

ex-

ists

a

K

such

that

for

all

$$k \geq$$

K

we

have

$$|A_k - a| <$$

$$\epsilon/2.$$

This

im-

plies

2.5.
 CON-
 VER-
 GENCE
 that
 the
 SESTS
 all
 $k \geq$
 K
 we
 have
 $|A_k -$
 $A_{k+1}| =$
 $|a_{k+1}| <$
 $\epsilon,$
 i.e.
 the
 se-
 quence
 can
 be
 shown
 to
 be
 even-
 tu-
 ally
 ar-
 bi-
 trar-
 ily
 close
 to
 zero.
 This
 is
 equiv-
 a-
 lent
 to
 the
 def-
 i-
 ni-
 tion
 for
 $a_n \rightarrow$
 0
 as
 $n \rightarrow$
 $\infty.$
 So
 this
 proves
 the
 sec-
 ond
 form
 of
 the

CHAPTER

2.

SERIES

the-

o-

rem.

□

But

note

that

a

se-

ries

can

still

di-

verge

even

if

the

se-

quence

of

terms

con-

verges

to

zero.

For

ex-

am-

ple

the

har-

monic

se-

ries,

$$\sum_{m=1}^{\infty} \frac{1}{m},$$

diverges

even

though

$$\frac{1}{m} \rightarrow$$

0

as

$$m \rightarrow$$

∞ .

The

di-

ver-

gence

of

the

har-

monic

se-

ries

2.5.
CON-
VER-
GENCE
cap
by
TESTS
shown

by
us-
ing
the
in-
te-
gral
test
(see
later)
as
well
as
the
ear-
lier
term-
grouping
ar-
gu-
ment.

A
ge-
o-
met-
ric
se-
ries
is,
as
the
name
sug-
gests,
the
sum
of
the
terms
of
a
ge-
o-
met-
ric
se-
quence.
The
con-
ver-
gence
of
a

CHAPTER

2.

SERIES

ge-
o-
met-
ric
se-
ries
can
be
de-
ter-
mined
us-
ing
the
fol-
low-
ing
test.

Theorem

2.4

(Ge-
o-
met-
ric
se-
ries
test).

The

ge-

o-

met-

ric

se-

ries

$\sum_{n=1}^{\infty} ar^{n-1}$

con-

verges

triv-

ially

if

$a =$

0.

If

$a \neq$

0

then

it

con-

verges

if

and

only

if

$|r| <$

1,

2.5.
CON-
VER-
GENCE
TESTS
case

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}.$$

Proof.

The
con-
ver-
gent
case

$a =$
0
is
clear.

So
we
as-
sume
that
 $a \neq$
0.

If
 $r =$
1
then
di-
ver-
gence
is
clear

as
the
se-
ries
is
just
 $a +$
 $a +$
 $a +$
 $a +$
....

So
as-
sume
 $r \neq$

1
and
let
 A_k
de-
note
the
par-
tial

CHAPTER

2.

SERIES

sum,

i.e.

$A_k =$

$\sum_{n=1}^k a_n.$

Most

of

the

terms

in

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fer-

ence

$A_k -$

rA_k

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$$A_k = \frac{a(1 - r^k)}{1 - r}.$$

Then

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CON-

VER-

GENCE

TESTS

$$\sum_{n=1}^{\infty} ar^{n-1} = \lim_{k \rightarrow \infty} A_k = \lim_{k \rightarrow \infty} \frac{a(1 - r^k)}{1 - r} = \frac{a(1 - \lim_{k \rightarrow \infty} r^k)}{1 - r}.$$

If

$r =$

-1

or

$|r| >$

1

then

r^k

di-

verges,

and

hence

so

does

the

se-

ries.

If

$|r| <$

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then

$r^k \rightarrow$

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$k \rightarrow$

∞ ,

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sum

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is

$a/(1 -$

$r)$,

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□

Theorem

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CHAPTER

2.

SERIES

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$\sum_{m=1}^{\infty} a_m$

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and

that

$a_m =$

$f(m)$

for

all

$m \geq$

1.

If

the

im-

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gral

$$\int_1^{\infty} f(t) dt = \lim_{T \rightarrow \infty} \int_1^T f(t) dt$$

exists

and

is

fi-

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then

the

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2.5.
CON-
VER-
GENCE
~~TESTS~~
~~TESTS~~
con-
verges.

If
the
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does
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then
the
orig-
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verges.

Proof.
We
sketch
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main
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proof
here
based
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[2.1.](#)
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CHAPTER

2.

~~SERIES~~

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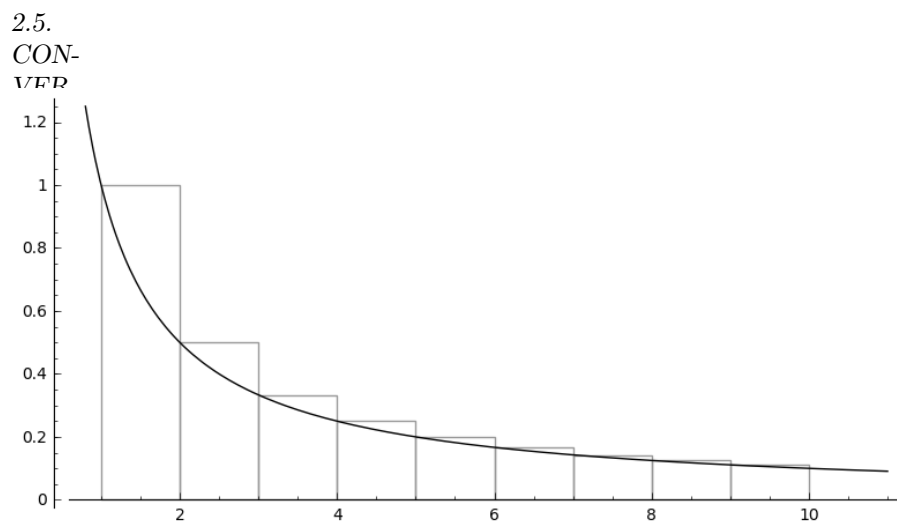


Figure 2.1:
The
in-
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and
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height
 $a_i =$
 $f(i)$.
In-
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that
 $\int_1^\infty f(t) dt$
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CHAPTER

2.

SERIES

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2.5. CON- VER- GENCE TESTS

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sions
then
we
have

Euler-Mascheroni constant

in the case, where the function f is
the reciprocal function $f(t) = 1/t$,
the quantity γ is known as the Euler-
Mascheroni constant, an important
quantity in analysis and number the-
ory where it occurs in many settings.
Note that it is still not known whether γ is irra-
tional or not, though most would bet
on that it is.

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CHAPTER

2.

SERIES

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CHAPTER

2.

SERIES

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Theorem

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$\sum_{m=1}^{\infty} a_m$

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m,

$a_m \geq$

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and

that

$\sum_{m=1}^{\infty} c_m$

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terms.

Then

•

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$m \geq$

1,

$a_m \leq$

c_m ,

then

$\sum_{m=1}^{\infty} a_m$
is

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$m \geq$

1,

$a_m \geq$

d_m ,

then

$\sum_{m=1}^{\infty} a_m$
is

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Good and bad behaviour

~~Proof~~ A good way to think about/remem-
ber this result is in terms of the good
and bad behaviour shown by a series.
We sketch

A convergent series $\sum c_m$ of positive
terms converges because the terms c_m
behave well (i.e. are small enough). If
 $a_m \leq c_m$ then the terms a_m are be-
having better and so $\sum a_m$ also con-
verges.

Similarly a divergent series $\sum d_m$ di-
verges because its terms behave badly
(i.e. they're too large), so if $a_m \geq d_m$
then the terms a_m are behaving worse,
and so $\sum a_m$ also diverges.

CHAPTER

2.

SERIES

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$$\sum_{m=1}^{\infty} c_m =$$

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$$m \geq$$

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have

$$a_m \leq$$

c_m .

Since

each

2.5.
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 $\sum_{m=1}^k a_m$
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CHAPTER

2.

SERIES

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$a_m \geq$

d_m .

Then

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$$\sum_{m=1}^k a_m \geq \sum_{m=1}^k d_m,$$

and
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 $\sum_{m=1}^k a_m$
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Theorem

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 $\sum_{m=1}^{\infty} a_m$
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terms,
i. e.
for*

CHAPTER

2.

SERIES

all

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$a_m \geq$

0,

and

that

$\sum_{m=1}^{\infty} c_m$

is

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terms

(i.e.

for

all

m ,

$c_m >$

0)

and

$\sum_{m=1}^{\infty} d_m$

is

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known

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of

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tive

terms.

Then

•

if

$\lim_{m \rightarrow \infty} \frac{a_m}{c_m} =$

L

and

$0 \leq$

$L <$

∞

then

$\sum_{m=1}^{\infty} a_m$

is

2.5.
CON-
VER-
GENCE^{con-}
TESTS^{ver-}
gent,

•

if
 $\lim_{m \rightarrow \infty} \frac{a_m}{d_m} =$

L

and

$0 <$

$L \leq$

∞

,

then

$\sum_{m=1}^{\infty} a_m$

is

di-

ver-

gent.

Good/bad behaviour

Again this result can be boiled
down to good/bad behaviour. If
this limit $\lim_{m \rightarrow \infty} a_m/x_m = L$ then *eventually*
limit a_m acts like a scalar multiple of x_m .
con- if $\sum x_m$ behaves well (or badly),
par- does $\sum a_m$.

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CHAPTER

2.

SERIES

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$$a_m/c_m$$

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tive
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M
such
that
for
all
 $m \geq$
 M

we
have
 $|a_m/c_m -$
 $L| <$

1,

hence

$$a_m/c_m <$$

$L +$

1

and

hence

$$a_m <$$

$$(L +$$

$$1)c_m.$$

There-
fore
the
stan-
dard
com-
par-
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son
test

2.5.
 CON-
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 GENCE
 says
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 ries
 $\sum_{m=M}^{\infty} a_m$
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 and
 hence
 so
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 ries
 $\sum_{m=1}^{\infty} a_m$.

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 a_m/d_m
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 M

CHAPTER

2.

SERIES

such

that

for

all

$m \geq$

M

we

have

$|a_m/d_m -$

$L| <$

$L/2,$

hence

$a_m/d_m >$

$L/2$

and

hence

$a_m >$

$(L/2)d_m.$

There-

fore

the

stan-

dard

com-

par-

i-

son

test

says

that

the

tail

se-

ries

$\sum_{m=M}^{\infty} a_m$

di-

verges

and

hence

so

does

the

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se-

ries

$\sum_{m=1}^{\infty} a_m.$

□

The

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2.5.
CON-
VER-
GENCE
~~OF~~
TESTS

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ries
or
a
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harmonic*
se-
ries

$$\sum_{m=1}^{\infty} \frac{1}{m^p}$$

for
some
 p .

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CHAPTER

2.

SERIES

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$$\sum_{m=1}^{\infty} \frac{(-1)^n}{m^p}$$

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$p >$

1,

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verges

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ally

when

$0 <$

$p \leq$

1

and

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wise.

Theorem

2.8

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test).

Con-

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$$\sum_{m=1}^{\infty} (-1)^m a_m,$$

where

the

2.5.
CON-
VER-
GENCE
TESTS

all
pos-
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tive
and
de-
creas-
ing,
i.e.
for
all
 $m \geq$
1,
 $a_m \geq$
 $a_{m+1} \geq$
0.
If
 $\lim_{m \rightarrow \infty} a_m =$
0
then
the
orig-
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ries
is
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ver-
gent.
But
if
 $\lim_{m \rightarrow \infty} a_m \neq$
0
then
the
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gent.

Proof.
We
give

CHAPTER

2.

SERIES

a

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main

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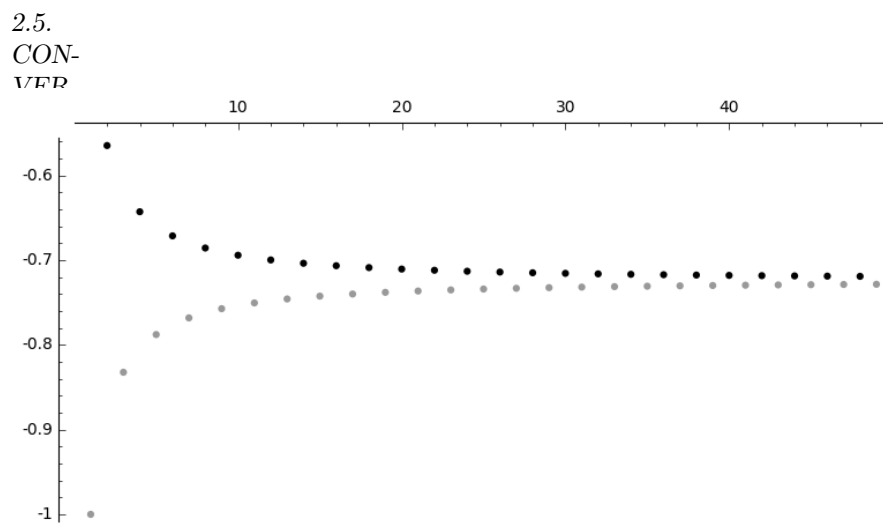


Figure 2.2:

The
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sums,
 A_k ,
of
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two
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se-
quences,
those
of
odd
in-
dex,
 $A_1, A_3, A_5, \dots$,
shown
in
grey,
and
those
of
even
in-
dex,
 $A_2, A_4, A_6, \dots$,
shown

CHAPTER

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~~SE~~RIES

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 is
 con-
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 gent,
 i.e.
 $A_{2k} \rightarrow$
 L_{even}
 and
 $A_{2k+1} \rightarrow$
 L_{odd}
 as
 $k \rightarrow$

CHAPTER

2.

SERIES

∞ .

But

since

the

step

size,

$|a_m|$,

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and

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zero

and

hence

$L_{\text{even}} =$

L_{odd}

and

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2.5.
CON-
VER-
GENCE
TESTS
□

**Theorem
2.9**

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*Con-
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the*

*se-
ries*
 $\sum_{m=1}^{\infty} a_m$
of

*pos-
i-
tive
terms,*

*i.e.
for*

all

m,

*$a_m >$
0.*

*Sup-
pose
that*

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quence*

*of
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*tios
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tive*

terms

is

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ver-*

gent

and

that

$$\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = L.$$

If

$$0 \leq$$

$$L <$$

1

then

the

CHAPTER

2.

SERIES

se-

ries

is

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gent.

If

$L >$

1

then

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Learning this result

This result boils down to the key idea:
 Suppose
 if $\lim_{m \rightarrow \infty} a_{m+1}/a_m = L$ then $\sum a_m$ is
 similar (compares to) to a geometric
 series with common ratio L . Hence
 $\sum a_m$ converges or diverges accord-
 ingly as $L < 1$ or $L > 1$.

the
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 rem
 and
 that

$$\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = L < 1.$$

Choose
 a

CHAPTER

2.

SERIES

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ϵ ,

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$\epsilon =$

$(1 -$

$L)/2$.

Then

there

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a

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M

such

that

for

all

$m \geq$

M

we

have

$|a_{m+1}/a_m -$

$L| <$

$(1 -$

$L)/2$,

i.e.

$a_{m+1}/a_m <$

$(L +$

$1)/2 <$

1.

Hence,

for

all

$m \geq$

M

we

have

$$a_{m+1} < \frac{L+1}{2} a_m.$$

Repeatedly

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lows

2.5.
 CON-
 VER-
 GENCE
 vs
 TESTS
 to
 say
 that
 for
 all
 $n \geq$
 1
 we
 have

$$a_{M+n} < \left(\frac{L+1}{2} \right)^n a_M.$$

Note
 that
 the
 se-
 ries
 $\sum_{n=1}^{\infty} \left(\frac{L+1}{2} \right)^n a_M$
 is
 a
 con-
 ver-
 gent
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 met-
 ric
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 ries,
 since
 $\frac{L+1}{2} <$
 1.
 So
 by
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 test
 the
 se-
 ries
 $\sum_{n=1}^{\infty} a_{M+n}$
 is
 also
 con-
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 gent.
 There-
 fore
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CHAPTER

2.

SERIES

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ries

$\sum_{m=1}^{\infty} a_m$

has

a

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ver-

gent

tail,

and

hence

is

con-

ver-

gent

as

re-

quired.

On

the

other

hand

is

we

sup-

pose

that

$$\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = L > 1,$$

then

we

fol-

low

a

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that

this

time

com-

pares

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tail

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our

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ries

to

a

2.5.
 CON-
 VER-
 GENCE
 di-
 VER-
 TESTS
 gent
 ge-
 o-
 met-
 ric
 se-
 ries
 $\sum_{n=1}^{\infty} \left(\frac{L+1}{2}\right)^n a_M,$
 for
 some
 M
 (re-
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 ber
 this
 time
 $(L+1)/2 >$
 $1)$.
 This
 im-
 plies
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 ries
 has
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 ver-
 gent
 tail
 and
 hence
 is
 it-
 self
 di-
 ver-
 gent,
 as
 re-
 quired.
 In
 the
 re-
 main-
 ing
 case
 where
 $\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = 1,$
 then
 try-

CHAPTER

2.

~~SE~~RIES

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above

ideas

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us

to

say

that

$\sum a_m$

has

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But

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sion

about

the

be-

haviour

2.5.
CON-
VER-
GENCE
TESTS



Theorem

2.10

(Root
test).

*Con-
sider
the*

*se-
ries*

$\sum_{m=1}^{\infty} a_m$
of

non-

*negative
terms,*

i.e.

for

all

m,

$a_m \geq$

$0.$

*Sup-
pose*

that

$\lim_{m \rightarrow \infty} (a_m)^{\frac{1}{m}} = L.$

If

$0 \leq$

$L <$

1

then

the

orig-

i-

nal

se-

ries

is

con-

ver-

gent,

if

$1 <$

$L \leq$

∞

then

the

se-

ries

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verges.

For

the

CHAPTER

2.

~~SE~~RIES

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main-

ing

case

$L =$

1

we

can

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draw

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sion

about

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ver-

gence.

Proof.

We

can

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$\sum a_m$

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L .

Hence

the

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appearance

of

the

$L <$

1

and

$L >$

2.6. POWER SERIES

1
cri-
te-
ria.
The
de-
tails
are
left
as
an
ex-
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cise.
□

Example 2.4.

We
shall
con-
sider
ex-
am-
ples
of
the
ap-
pli-
ca-
tions
of
these
tests
in
the
lec-
tures.

2.6 Power series

Using
se-
ries
we
can
de-
fine
a
class
of
func-
tions
called

CHAPTER

2.

~~S~~ERIES

power

se-

ries.

We

shall

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at

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topic

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we

are

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lus.

Definition

2.5.

A

se-

quence

of

terms

$\{a_m\}_{m=1}^{\infty}$

de-

fines

a

power

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ries

func-

tion

A

by

the

for-

2.6.

POWER

SERIES

mula

$$A(x) = \sum_{m=1}^{\infty} a_m x^m.$$

Of

course

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mula

for

$A(x)$

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For

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We

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CHAPTER

2.

SERIES

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test.

According

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test
the
se-
ries
for-
mula
for
 $A(x)$
will
con-
verge
when

$$\lim_{m \rightarrow \infty} \left| \frac{a_{m+1}x^{m+1}}{a_mx^m} \right| < 1,$$

which
is
equiv-
a-
lent,
af-
ter
sim-
pli-
fy-
ing,
to

$$|x| < \lim_{m \rightarrow \infty} \left| \frac{a_m}{a_{m+1}} \right|.$$

Definition

2.6.

The
quan-
tity
 R
de-
fined
by

$$R = \lim_{m \rightarrow \infty} \left| \frac{a_m}{a_{m+1}} \right|,$$

is
known

2.6.
 POWER
~~SERIES~~
 as
 the
ra-
dious
of
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ver-
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 of
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 se-
 ries
 $\sum_{m=1}^{\infty} a_m x^m$.
 The
 se-
 ries
 con-
 verges
 if
 $|x| <$
 R
 and
 di-
 verges
 if
 $|x| >$
 R .

Note
 that
 the
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 does
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 about
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 ries
 when
 $|x| =$
 R .
 That
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CHAPTER

2.

SERIES

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Theorem

2.11

(Tay-
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*Sup-
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$[a, x]$

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ated

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1

times,

2.6.
POWER
SERIES

for
some
 $n \geq$
0.

Then
there
ex-

ists
a
num-

ber
 $c \in$
 (a, x)
such
that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n,$$

where
 R_n
is
the
La-
grange
re-
main-
der
and

is
given
by

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Taylor's
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CHAPTER

2.

SERIES

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x

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f

about

the

point

a :

$$f(a+x) = f(a) + f'(a)x + \frac{f''(a)}{2!}x^2 + \cdots + \frac{f^{(n)}(a)}{n!}x^n + R_n.$$

It

is

of-

ten

the

case

that

$R_n(x) \rightarrow$

0

as

2.6.
POWER
SERIES
 $n \rightarrow$

∞
and
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may
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Tay-
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sion
about
the
point
 a :

$$f(a+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} x^n.$$

Two
im-
por-
tant
things
to
note
are
that
for
all
 $x \in$
 $\mathbb{R}$
we
have

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

and
thatn
if
 $-1 <$
 $x <$
1
then

$$\lim_{n \rightarrow \infty} x^n = 0.$$

CHAPTER

2.

SERIES

The

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cial
case
where
we
ex-
pand
a
func-
tion
about
the
point
 $a =$
 0
gives
rise
to
what
is
called
a
*Maclau-
rin*
*se-
ries*.

Definition

2.7.

The
Tay-
lor
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ries
ex-
pan-
sion
of
a
func-
tion
 f
about
the
point
 0
is
called
the
*Maclau-
rin*
*se-
ries*
of

2.6.
POWER
~~*SERIES*~~
f,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

We
shall
re-
turn
to
the
topic
of
Tay-
lor
se-
ries
af-
ter
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have
stud-
ied
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tion
in
more
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tail.

In
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[2.1](#)
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will
see
Maclau-
rin
se-
ries
of
some
stan-
dard
func-
tions.

CHAPTER

2.

SERIES.

Maclarin series	Interval of convergence
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$	$(-\infty, \infty)$
$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$	$(-\infty, \infty)$
$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$	$(-\infty, \infty)$
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$	$(-1, 1)$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{n}$	$(-1, 1)$

Table 2.1:

Some
stan-
dard
Maclau-
rin
se-
ries

Exercises

2.1.

These
prob-
lems
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quire
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and
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sults
from
this
chap-
ter.
We

2.6.
POWER
~~*SERIES*~~
 shall
 use
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1. Investigate
 the
 con-
 ver-
 gence
 of
 the
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 ing
 se-
 ries,

a)

$$\sum_{n=1}^{\infty} (2n-1) = 1+3+5+7+9+\dots,$$

b)

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\dots,$$

c)

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+\dots$$

2. Using
 a
 di-
 a-
 gram

CHAPTER

2.

SERIES

of
the
num-
ber
line
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ves-
ti-
gate
the
con-
ver-
gence
of
the
fol-
low-
ing
se-
ries,

$$(a) \sum_{n=1}^{\infty} (-1)^n, \quad (b) \sum_{n=1}^{\infty} (-1)^n \left(\frac{n+1}{n+2} \right).$$

3.

The
ge-
o-
met-
ric
se-
ries
 $\sum_{n=1}^{\infty} ar^{n-1}$
has
par-
tial
sums
 S_k
given
by

$$S_k = \sum_{n=1}^k ar^{n-1} = \frac{a(1-r^k)}{1-r}.$$

Use
this
re-
sult
to
in-
ves-
ti-
gate
the

2.6.

POWER

~~S~~ERIES

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ver-

gence

of

the

se-

ries

$\sum_{n=1}^{\infty} nar^{n-1}$.

Hint:

Let

T_k

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k^{th}

par-

tial

sum

of

this

se-

ries.

Con-

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the

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tity

$(1 -$

$r)T_k \dots$

4.

Recall

from

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ture

that

the

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ric

se-

ries

$\sum_{n=1}^{\infty} ar^{n-1}$

has

par-

tial

sums

S_k

given

by

$$S_k = \sum_{n=1}^k ar^{n-1} = \frac{a(1-r^k)}{1-r}.$$

Therefore

CHAPTER

2.

SERIES

the
se-
ries
is
con-
ver-
gent
only
when
 $|r| <$
1,
in
which
case
the
in-
fi-
nite
se-
ries
sums
to
 $\frac{a}{1-r}$.

For
the
fol-
low-
ing
def-
i-
ni-
tions
of
 x_n
de-
ter-
mine
whether
the
se-
ries
 $\sum_{n=1}^{\infty} x_n$
con-
verges
and
if
so
find
the
sum
of
the
se-
ries.

a)

2.6.

POWER

SERIES

$$x_n = \frac{2}{3} \left(\frac{1}{2}\right)^n - \frac{1}{4} \left(\frac{2}{3}\right)^n.$$

b)

$$x_n = \left(\frac{3}{2}\right)^{2n} - \frac{1}{2} \left(\frac{2}{3}\right)^{n+1}$$

5.

Consider

the

se-

ries

$$\sum_{n=1}^{\infty} \frac{2}{n^2 + 2n}.$$

Show

that

this

is

a

con-

ver-

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se-

ries

by

find-

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the

par-

tial

frac-

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ex-

pan-

sion

of

$$\frac{2}{n^2 + 2n}$$

and

ex-

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the

k^{th}

par-

tial

sum

S_k .

6.

Consider

the

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CHAPTER

2.

~~96~~ SERIES

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sion

$2.104104104104\dots$

By
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by
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sion.

7.

Determine
whether
the
fol-
low-
ing
se-
ries
is
con-
ver-
gent

2.6.

POWER

~~S~~ERIES

or

di-

ver-

gent

$$\sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)}.$$

8.

Determine

whether

the

fol-

low-

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se-

ries

are

con-

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gent

or

di-

ver-

gent

a)

$$\sum_{n=1}^{\infty} \frac{2}{n!},$$

b)

$$\sum_{n=1}^{\infty} \frac{n!}{n^n},$$

c)

$$\sum_{n=1}^{\infty} \left(\frac{2n-1}{3+4n} \right)^n,$$

d)

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}.$$

9.

Determine

whether

the

fol-

low-

ing

se-

ries

CHAPTER

2.

~~SE~~RIES

con-
verges
or
di-
verges:

$$\sum_{n=1}^{\infty} a_n, \quad \text{where } a_n = \frac{2n^2 + 4}{3n^3 - 2n}.$$

10.

Determine
whether
the
fol-
low-
ing
se-
ries
con-
verges
or
di-
verges:

$$\sum_{n=1}^{\infty} b_n \text{ where } b_n = \frac{2n + 5}{\sqrt{n^6 + n}}.$$

11.

Determine
whether
the
fol-
low-
ing
se-
ries
con-
verges
or
di-
verges:

$$\sum_{n=1}^{\infty} c_n \text{ where } c_n = \frac{n!}{5^{n+2}}.$$

12.

Determine
whether
the
fol-
low-
ing
se-
ries

2.6.

POWER

~~SERIES~~

con-

verges

or

di-

verges:

$$\sum_{n=1}^{\infty} d_n \text{ where } d_n = \frac{n^2 2^{n+5}}{2^{2n}}.$$

13.

Determine

whether

the

fol-

low-

ing

se-

ries

con-

verges

or

di-

verges:

$$\sum_{n=1}^{\infty} e_n \text{ where } e_n = \frac{\sin^2(n)}{n(n+3)}.$$

14.

Determine

whether

the

fol-

low-

ing

se-

ries

con-

verges

or

di-

verges:

$$\sum_{n=1}^{\infty} f_n \text{ where } f_n = \frac{\sqrt{n+1}}{n\sqrt{n+4}}.$$

15.

Determine

whether

the

fol-

low-

ing

se-

ries

con-

verges

CHAPTER

2.

~~SERIES~~

or
di-
verges:

$$\sum_{n=1}^{\infty} g_n \text{ where } g_n = \frac{n2^n}{n!}.$$

16.

Write
down
the
MacLau-
rin
se-
ries
(i.e.
Tay-
lor
se-
ries
about
 $x =$
0)
for
the
ex-
po-
nen-
tial
func-
tion,
 e^x .

Hence
find
the
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of
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cor-
rect
to
four
dec-
i-
mal
places.

17.

Find

2.6.

POWER

~~SERIES~~

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the
ap-
prox-
i-
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tion

$$\sin x \approx x - \frac{x^3}{6},$$

given
that
 $|x| < \frac{1}{2}.$

18.

Approximate
 $\sin \frac{\pi}{5}$
to
four
dec-
i-
mal
places
us-
ing
an
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ate
Tay-
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